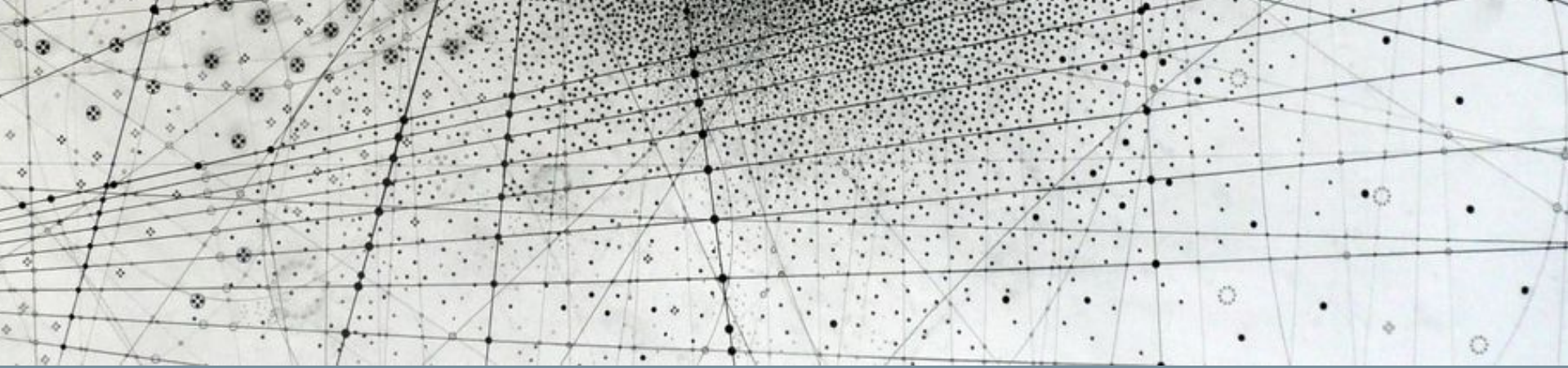




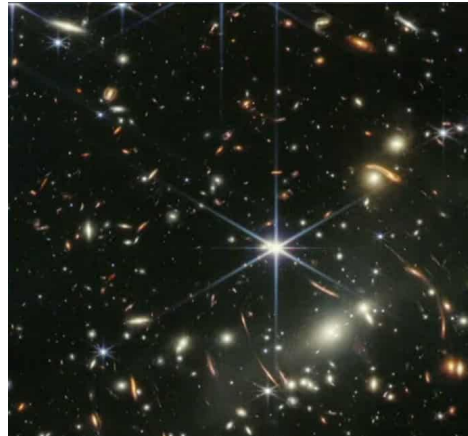
Research Methods in Machine Learning & Physics

Quick recap

Where in physics do we expect to find foundation models being used?

- Data-rich
- Multi-modal / highly specialized data
 - Existing foundation models are insufficient to describe the data
- Already comfortable with high-performance computing work
- Unsolved yet easily-verifiable tasks
- Generic or nearly-universal phenomena
 - e.g. objects in space that can be measured with multiple telescopes, identical fundamental particles, fluid dynamics across a variety of scales
- Funding and societal preferences/priorities

Where in physics do we expect to find foundation models being used?



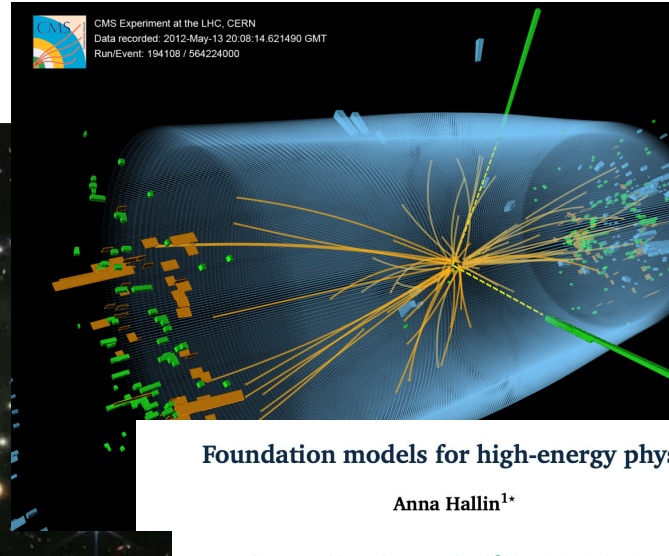
<https://arxiv.org/pdf/2510.17960>

Polymathic

AION-1: Omnimodal Foundation Model for Astronomical Sciences

Liam Parker^{1,2,3,4}, Francois Lanusse^{5,2}, Jeff Shen⁶, Ollie Liu⁷, Tom Hehir⁸, Leopoldo Sarra³, Lucas Meyer³, Micah Bowles⁹, Sebastian Wagner-Carena^{2,3}, Helen Qu², Slavash Golkar^{2,3}, Alberto Bietti³, Hatim Bourfoune¹⁰, Nathan Cassereau¹⁰, Pierre Cornette¹⁰, Keiya Hirashima^{2,11}, Geraud Krawezik², Ruben Ohana², Nicholas Lourie², Michael McCabe^{2,3}, Rudy Morel², Payel Mukhopadhyay^{1,8}, Mariel Pettee¹², Bruno Regaldo-Saint Blancard², Kyunghyun Cho³, Miles Cranmer⁸, Shirley Ho^{2,3,6}

¹University of California, Berkeley, ²Platoin Institute, ³New York University, ⁴Lawrence Berkeley National Laboratory, ⁵Université Paris-Saclay, Université Paris Cité, CEA, CNRS, AIM, ⁶Princeton University, ⁷University of Southern California, ⁸University of Cambridge, ⁹University of Oxford, ¹⁰IDRIS, CNRS, ¹¹RIKEN Center for iTHEMS, ¹²University of Wisconsin-Madison



Foundation models for high-energy physics

Anna Hallin^{1*}

<https://arxiv.org/pdf/2509.21434>

<https://arxiv.org/pdf/2509.13805>

TOWARDS A PHYSICS FOUNDATION MODEL

Florian Wiesner

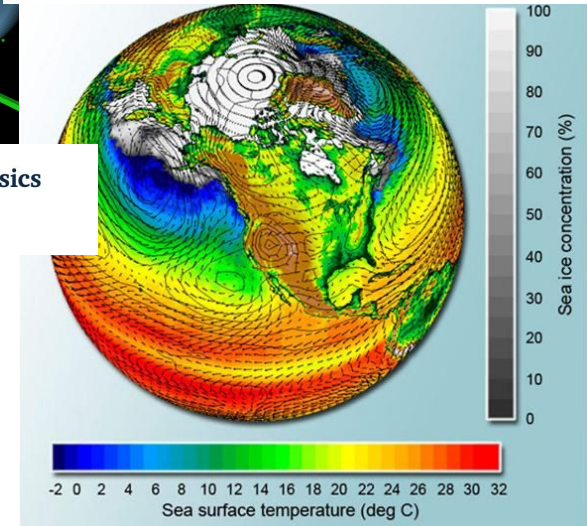
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"AI scientists"



Building the AI Physicist

FirstPrinciples is building the AI Physicist, a domain-specialized system designed to understand the nature of reality.

[Learn More ↗](#)

The AI Scientist: Towards Fully Automated Open-Ended Scientific Discovery

Chris Lu^{1,2,*}, Cong Lu^{3,4,*}, Robert Tjarko Lange^{1,*}, Jakob Foerster^{2,†}, Jeff Clune^{3,4,5,†} and David Ha^{1,†}

^{*}Equal Contribution, ¹Sakana AI, ²FLAIR, University of Oxford, ³University of British Columbia, ⁴Vector Institute, ⁵Canada CIFAR AI Chair, [†]Equal Advising

sakana.ai

The AI Scientist: Towards Fully Automated Open-Ended Scientific Discovery

August 13, 2024



periodic labs

ACCELERATE SCIENCE

Today, we introduce Periodic Labs. Our goal is to create an AI scientist.

Kosmos: An AI Scientist for Autonomous Discovery

Ludovico Mitchener^{*,1,†}, Angela Yiu^{*,1}, Benjamin Chang^{*,1,2}, Mathieu Bourdenx^{3,4,5}, Tyler Nadolski¹, Arvis Sulovari¹, Eric C. Landsness^{5,6}, Dániel L. Barabási^{7,8}, Siddharth Narayanan¹, Nicky Evans⁹, Shriya Reddy¹⁰, Martha Foiani^{3,4}, Aizad Kamal⁶, Leah P. Shriver^{11,12,13}, Fang Cao¹⁰, Asmamaw T. Wassie¹, Jon M. Laurent¹, Edwin Melville-Green¹, Mayk Caldas¹, Albert Bou¹, Kaleigh F. Roberts¹⁴, Sladjana Zagorac¹⁵, Timothy C. Orr⁶, Miranda E. Orr^{6,16}, Kevin J. Zwezdaryk^{17,18,19}, Ali E. Ghareeb¹, Laurie McCoy¹, Bruna Gomes¹⁰, Euan A. Ashley¹⁰, Karen E. Duff^{3,4,5}, Tonio Buonassisi^{9,20}, Tom Rainforth², Randall J. Bateman^{5,6}, Michael Skarlinski¹, Samuel G. Rodrigues^{1,7,‡}, Michaela M. Hinks^{1,†}, Andrew D. White^{1,7,‡}

<https://arxiv.org/pdf/2511.02824>

What are physicists for?

- Making discoveries
- Collecting data
- Analyzing data
- Writing papers
- Giving talks
- Mentoring / teaching students

What does it mean to “accelerate science”?

Is this something we should aim for?

Why should human scientists exist?

How to give a good academic talk

Take 3 minutes to write down some thoughts:

- Think about a really great academic talk you've seen.
 - Who was the speaker & what was the setting?
 - What do you remember about the talk?
 - What made it effective?
- What lessons have you already learned about how to give a good talk?

□ HOW TO START

□ 4 SAMPLES

□ THE TOOLS
TIME & PLACE
WORDS & PROPS
PROJECTIONS

$$Q_{\text{Quality}} = f(K, P, \dots)$$



Building a present system

usually something like: $\text{present} = \text{past} + \text{future}$
and is often the past and future are past

Disagreement to do the future column and

Present

the past and future

the past and future

the past and future

Future

the past and future

the past and future

the past and future

usually, $\text{present} = \text{past} + \text{future}$ $\text{present} = \text{past}$

the past and future

The big picture

- **Know your mission**
 - Is this a conference? Seminar? Colloquium?
 - What do the attendees already know, and why did they come?
 - Who is your audience / who are you trying to impress?
- **Examples:**
 - You want as many people as possible to read & use & cite your latest paper
 - You want to get closer to your peers in a new subfield
 - You're looking for potential collaborators
 - You want to convince a search committee that you have a strong vision for the field
 - This talk will mostly be looked at later in PDF form

Communicating Ideas Rather than Research

Giving a talk is a lot like selling a car.

Consider the used car salesperson. When you say you are interested in “that one,” they do not read you the owner’s manual. They do not tell you what each light on the dashboard means. No. They let you kick the tires, they take you for a test drive, they let you ask questions. They want you to discover that the car is great, and you can figure out what all the dashboard lights mean on your own time—*after you buy*.

In the same way, a talk should not be an exhaustive (or condensed) reading of the paper. A talk should be an advertisement—for the paper and for you as a productive, competent scholar. Your goal is not to communicate all the nuances of the research, but rather, to sell them on the idea of the paper—it’s theoretical importance, its exciting research design, its novel data—such that they dig into those nuances *on their own time*. Take the audience for a test drive, don’t talk to them about the tire pressure.

How to speak

- Actually use the microphone!
 - Be careful of turning your head / looking at the screen
- Look your audience in the eyes
 - **Important trick:** memorize your first couple of sentences
- Speak loudly and clearly
- Be aware of your pacing
 - You will likely speed up by ~20% due to nerves – make sure to include pauses
 - Time yourself!

How to start your talk

- **Introduce yourself**

- "Hi everyone, it's great to be here. My name is Mariel, and I'm an assistant professor of Physics at UW-Madison."

- **Give some framing**

- "Lately, I've been interested in understanding how to build foundation models for physics data, but I've found that there aren't many guidelines for how to construct foundation models with data from diverse areas."

- **State your promise to the audience (why should they listen?)**

- "So, today, I'd like to share with you my group's latest paper on constructing multimodal datasets with controllable mutual information. This talk will explore how to view self-supervised learning through an information theoretic lens, and hopefully might spark some interesting discussions about how to build interdisciplinary foundation models in the future."

The medium = the message

- Slides are designed to sell things
 - (You could be selling ideas, seeking wider exposure)
 - This isn't necessarily a bad thing – just be aware of it
- Chalk and/or long-form writing is more useful for informing
 - These slow you down – and that's the point!
 - Slides can go very quickly and hinder understanding

The medium = the message

I once asked a junior professor how—given the need to publish, teach, and meet with students—he had time to keep up with all of the new scholarship coming out each month. His response: he didn't. He said he kept up with new work primarily by attending talks, workshops, and conferences.

I share this long-winded windup that has little to do with talks for a reason—because this was the moment I realized that if you want others to read your work, you cannot simply publish it and assume others will find (and cite) it. *You need to sell it.*

– *A Long Guide to Giving a Short Academic Talk*, <https://benjamin noble.org/blog/short-talk>

The medium = the message

PRODUCTIVITY

Jeff Bezos Says This Is Why He Banned PowerPoint at Amazon. It's a Lesson in How to Think

It certainly makes Amazon's meetings 'unusual,' says Bezos. But it also makes them better.

EXPERT OPINION BY JUSTIN BARISO, AUTHOR, EQ APPLIED @JUSTINJBARISO

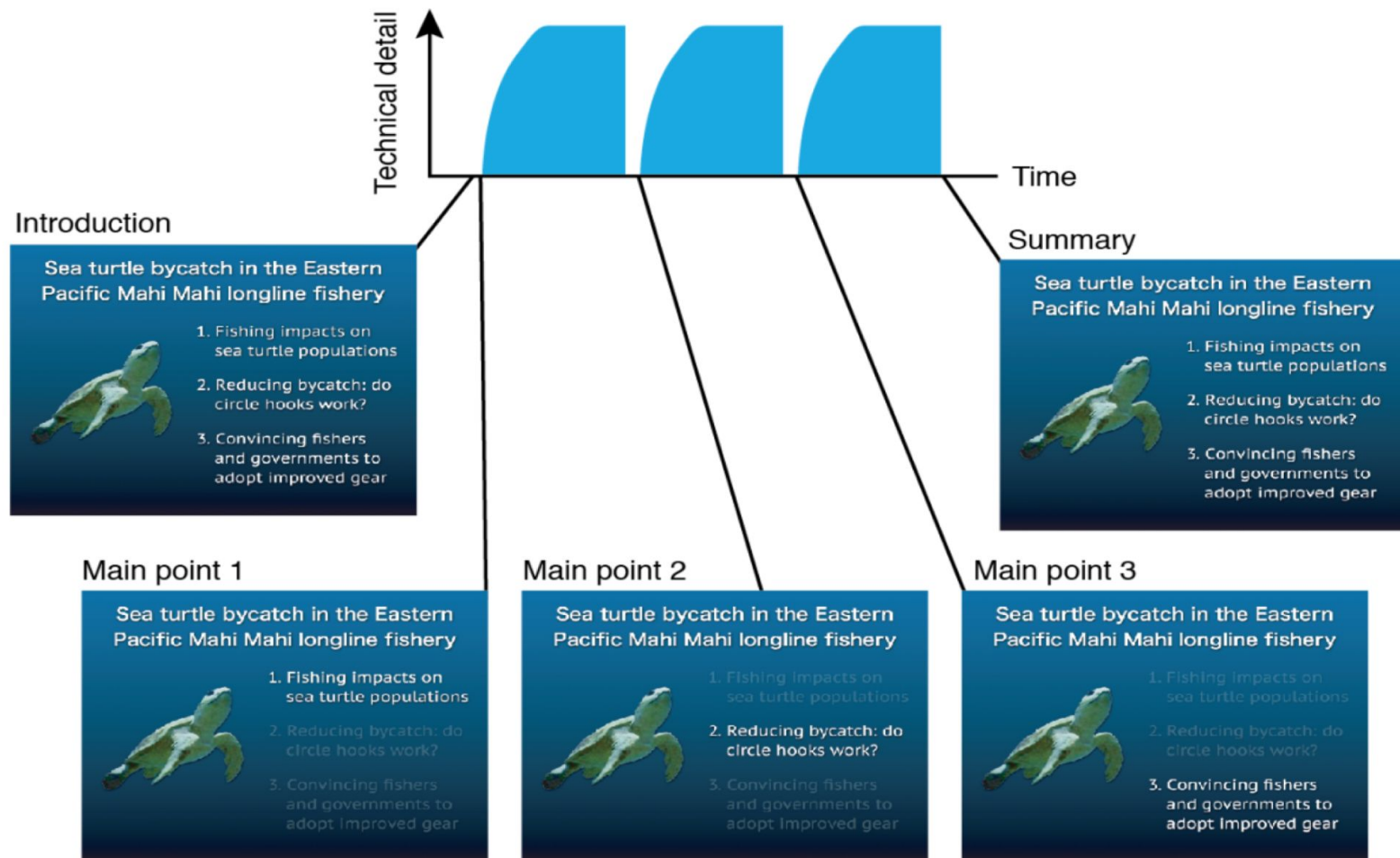
DEC 20, 2023

Another major problem of PowerPoint, says Bezos, is that it's designed to persuade, to get others to feel—*not* to get them to think. “It’s kind of a sales tool,” Bezos says.

“The other problem with PowerPoint,” says Bezos, “is it’s easy for the author and hard for the audience.”

Designing your slides

- **Only use a table of contents if you will speak for >30 minutes**
- **Make sure your fonts are large enough**
- **Use high-contrast colors**
- **Don't use a laser pointer**
 - Instead, use arrows that point to important parts of your plots



Designing your slides

- **Only include a plot** if you can show it at full-size (i.e. you can read the axis labels easily) and **if you are going to explain it** during your talk
 - Always describe the x and y axes, and say how to “read” the plot
 - You *will* get asked about any striking parts of a plot – make sure you know it inside & out.
- **Don’t overload your slides with words!**
 - 1 slide = 1 idea
 - **Tip:** Use a full sentence instead of a title

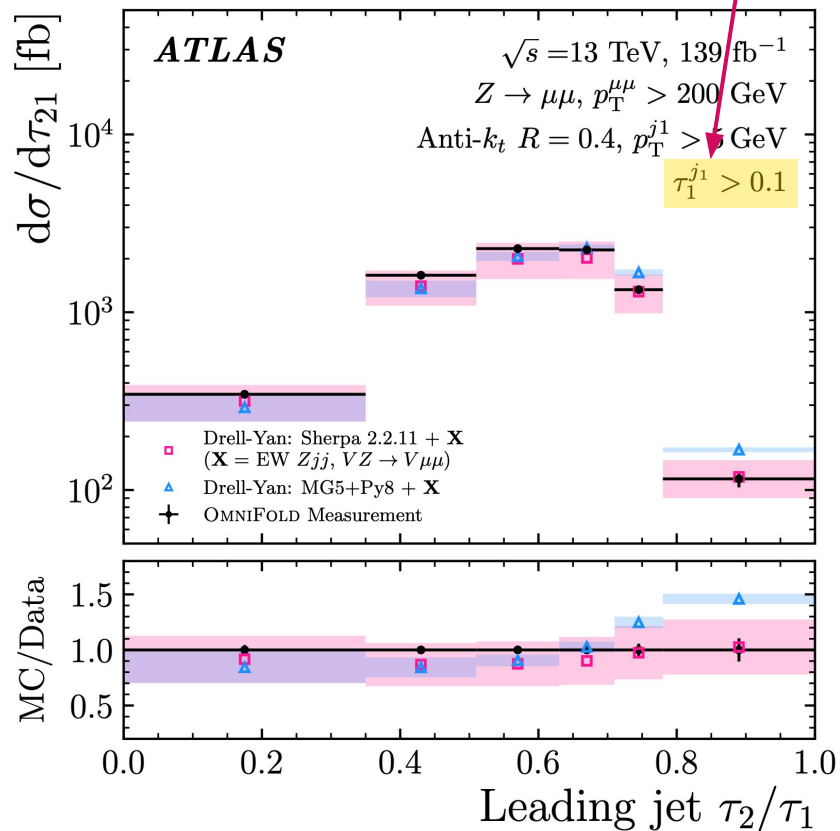
Each slide should only convey one main idea.

Overloading your slides with words is distracting.

Each slide should only convey one main idea.

IRC-safe phase space!

- **Don't overload your slides with words and formulas!**
 - 1 slide = 1 idea
 - **Tip:** Use a full sentence instead of a title
- Also, **use animations** to make your content more digestible



□ HOW TO START

□ 4 SAMPLES

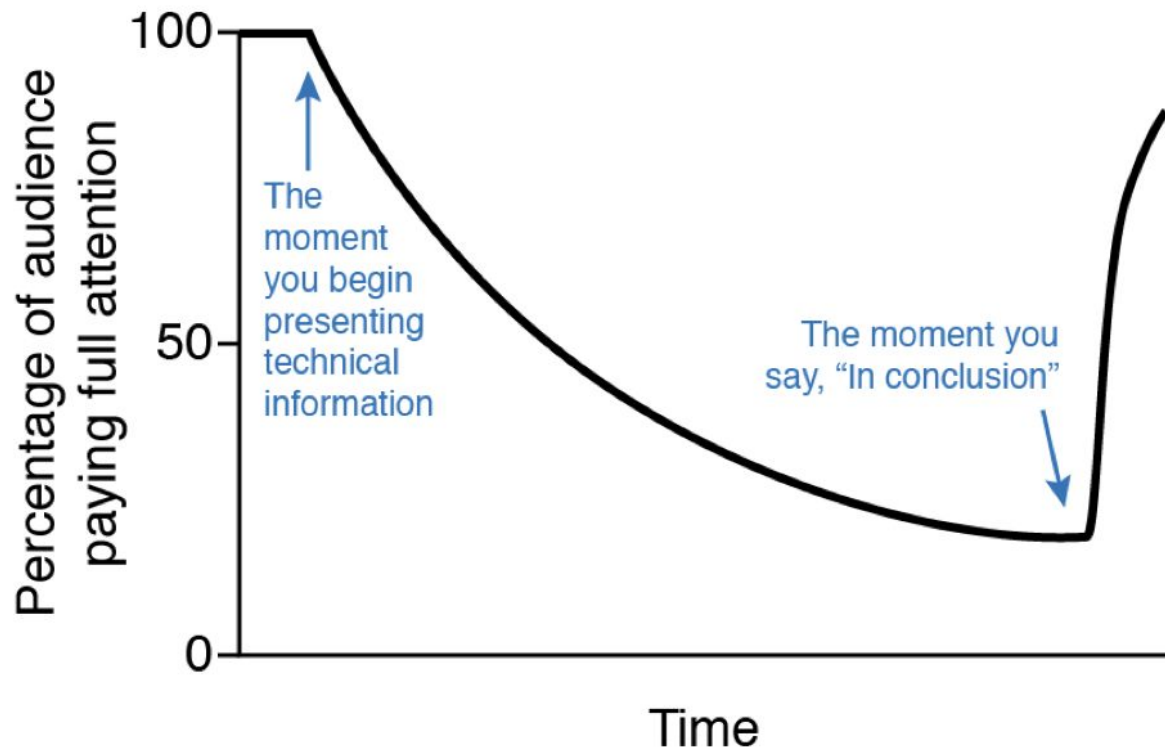
□ THE TOOLS
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$$Q_{\text{Quality}} = f(K, P, \dots)$$

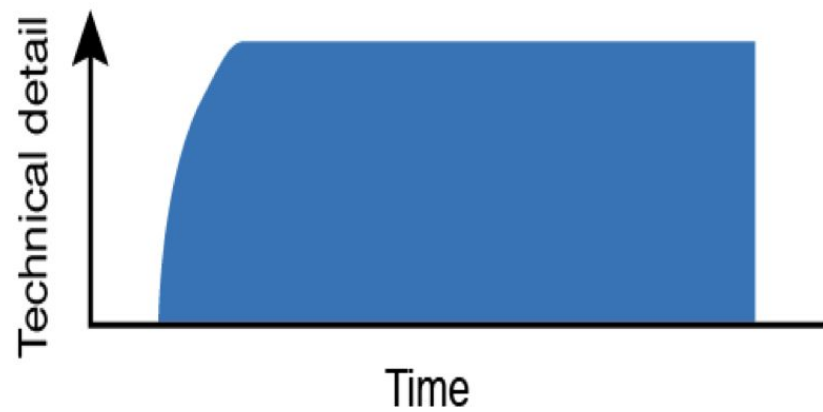


Other miscellaneous tips

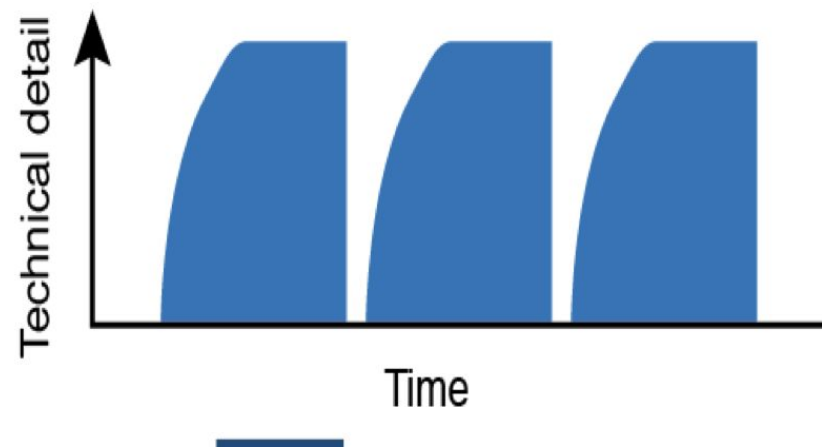
- **What is your message?**
 - Play "Jeopardy": what is the question to which your message is the answer?
 - Use these questions to construct your framing slides
- **Build in some healthy repetition** – people will not be paying consistent attention
- **"Verbal punctuation": provide landmarks to get people back into your talk**
 - Put in some breaks every ~10 minutes
 - Summarize what you just said, and then introduce your next point
 - Another way to get people back in: ask a question



Before



After



Other miscellaneous tips

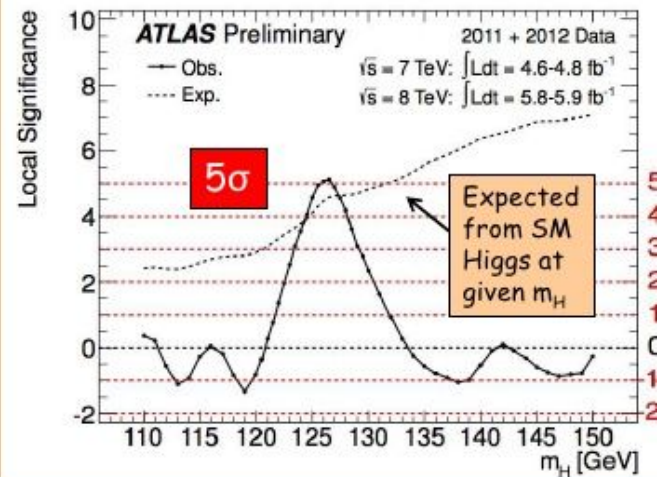
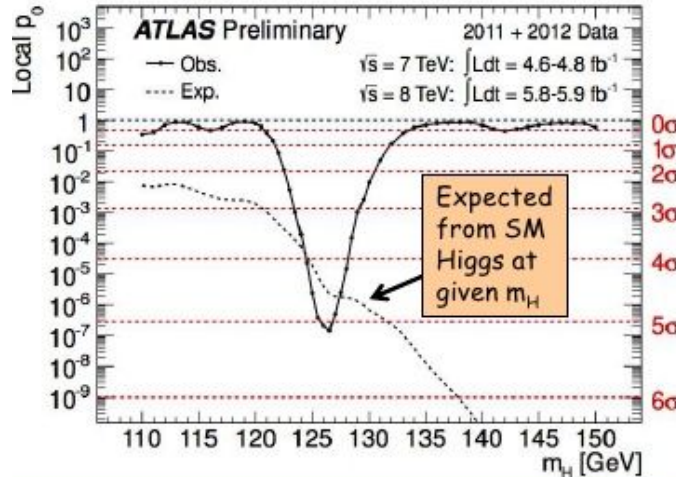
- **Rule of thumb: 1 slide = 1 minute**
- **If you're running behind...**
 - Don't say: "I'll skip this slide due to time, but essentially the point here is... [explains the slide anyways]"
- **Always add a summary slide with just a handful of key points.**
 - Rehearse your ending statement
- **In the Q&A, it's a good idea to restate the question.** (It's okay to say "I don't know"!)

Other miscellaneous tips

- **Let your enthusiasm show!**
- **This is an art form, at the end of the day.**
 - And ideally, it helps you appreciate the full context and meaning of your work.

Examples

Combined results: the excess



Maximum excess observed at

$m_H = 126.5 \text{ GeV}$

Local significance (including energy-scale systematics)

5.0σ

Probability of background up-fluctuation

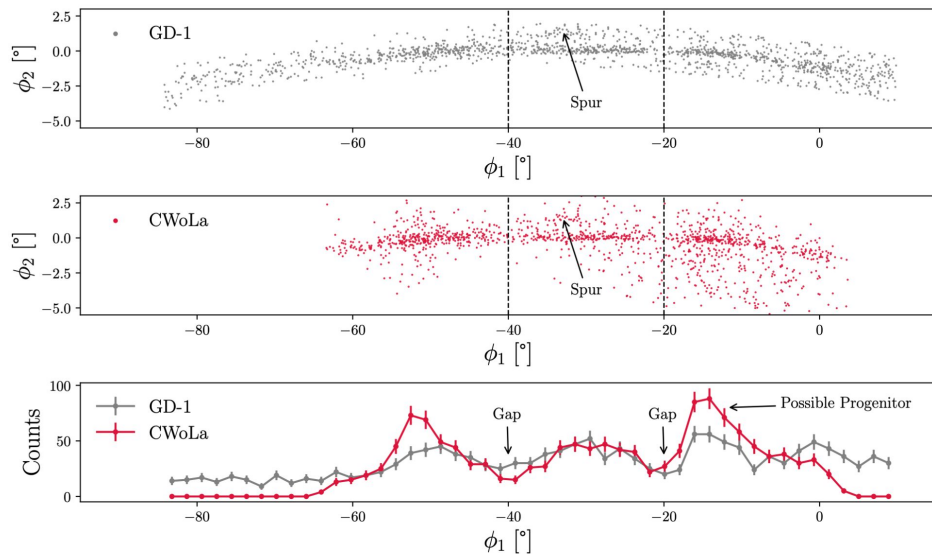
3×10^{-7}

Expected from SM Higgs $m_H=126.5$

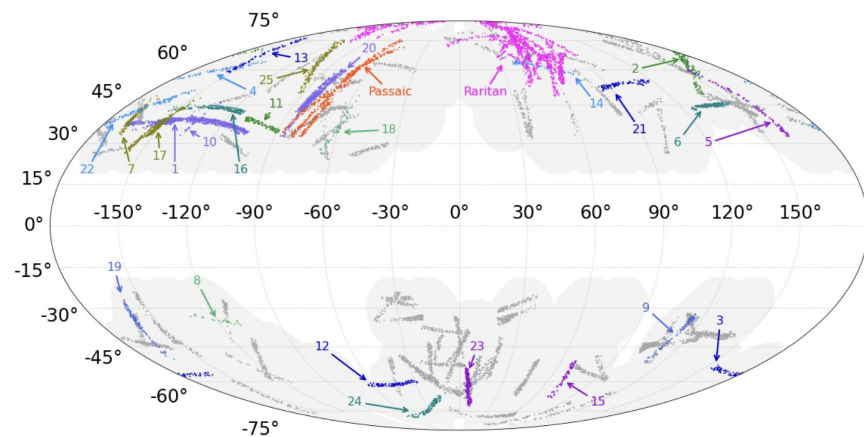
4.6σ

Global significance: 4.1-4.3 σ (for LEE over 110-600 or 110-150 GeV)

State-of-the-art **HEP anomaly detection** relies on finding group anomalies.
 (and we can apply this specialized expertise to resonances in other domains!)



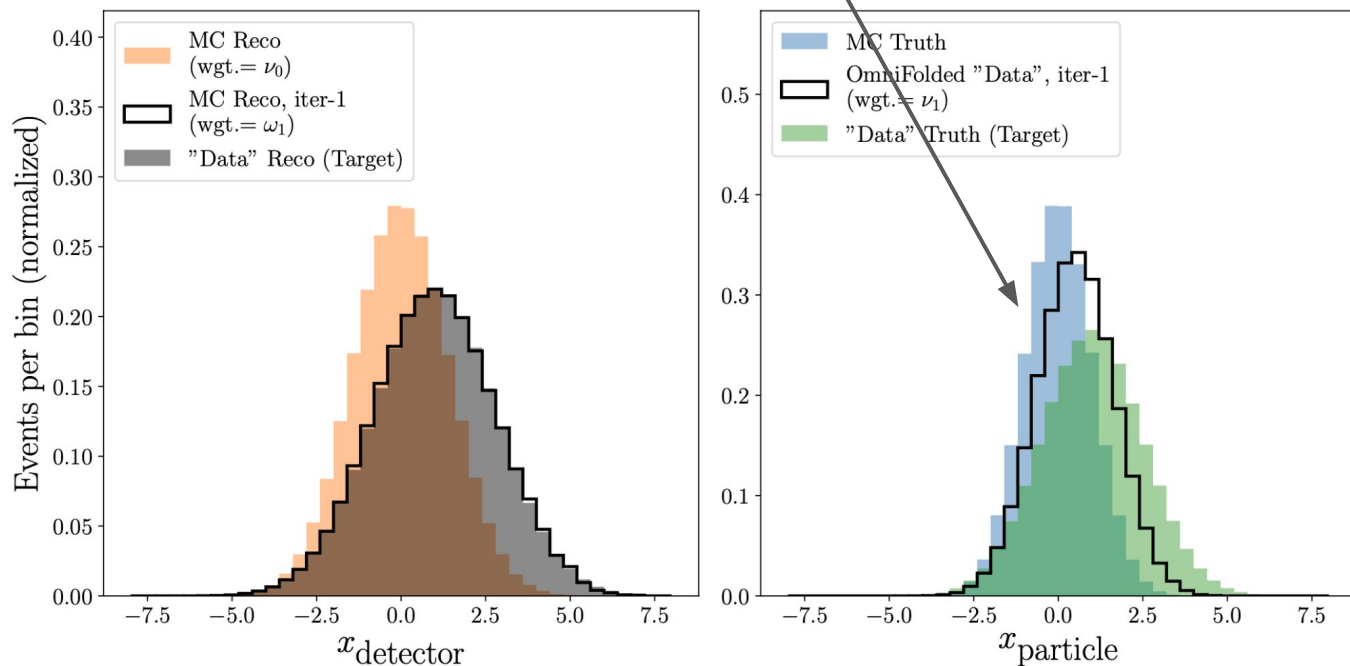
arXiv:2305.03761



arXiv:2509.08064

OmniFold reweights truth-level MC to estimate “truth-level” data.

After one iteration of our method, the **truth-level MC** is **reweighted** to look more like **truth-level data**.



Resources:

- [A Long Guide to Giving a Short Academic Talk](#)
- [Tips for Conference Presenting](#)
- [How to Speak](#)

In-class work

With a partner:

- Set a timer for 3 minutes and explain a recent research project to your partner.
 - (You can use one of your class projects, if you like, but ideally it should be related to your own research.)
 - The listener then has to summarize what they understood.
 - Take note of what information was or was not conveyed
 - Then, switch roles
- Next, spend some time writing down the first 5 sentences of a talk on this research. Make sure you include:
 - Introducing yourself
 - Some framing / context
 - What you're going to show
 - Why it matters / why the audience should care
- Then, memorize it and prepare to present it to the class